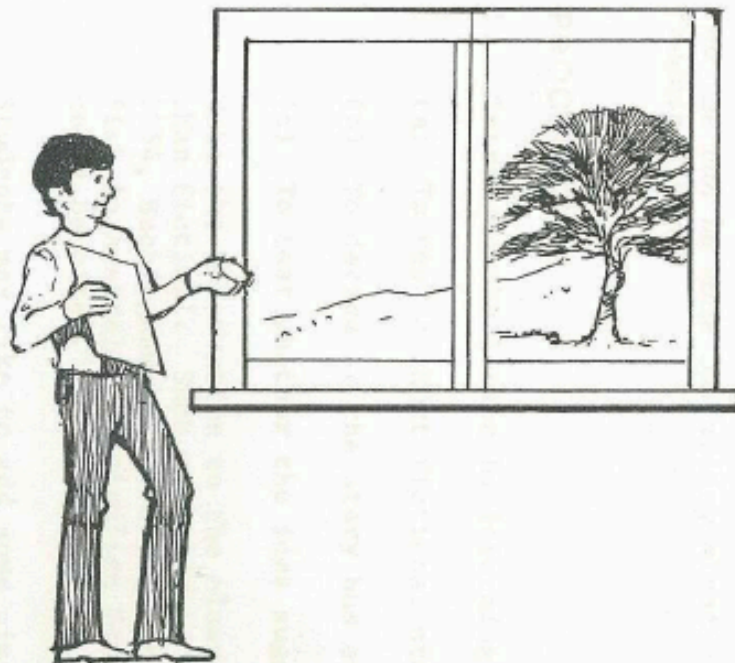
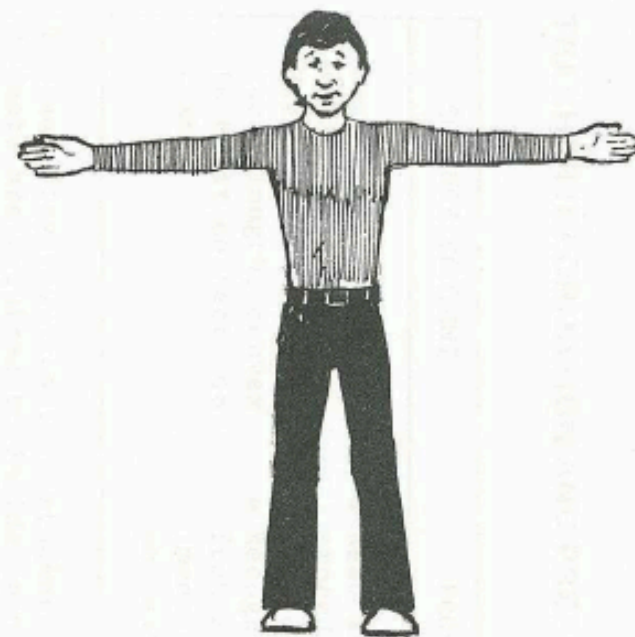




LENSES AND IMAGES



PROCEDURE 1

Stand 2 to 3 metres from a window and hold the sheet of paper and a convex lens as shown in the diagram.

Move the sheet backwards and forwards to see if you can produce an image on the page.

Carefully record your observations in a table.

Try the procedure again, using a concave lens.

You could try another lens of a different thickness and compare the distances between the sheet of paper and the lens.

PROCEDURE 2

Ask a member of your group to stand approximately 10 metres away from you. Holding the convex lens at arm's length, record the appearance of your partner, as seen through the lens. Then ask your partner to carry out the following:

- (a) Move his or her hands upwards.
- (b) Move the left then right hand.

When you are recording your observations, use your left and right hand when describing which way the image moves.

Now ask your partner to walk towards you, stopping 10 cm from the lens. Carefully record your observations.

Repeat this procedure, using a concave lens.

ANSWER THESE QUESTIONS

1. The eye has a convex lens. What would the images produced by the lens look like?
2. What does the brain do to the images it receives from the retina?
3. Explain why only convex lenses are used in cameras, binoculars and slide projectors.
4. The glass lens has a fixed focal length. Suggest how cameras overcome this problem. Explain what happens when you change the distance between the camera and the object.